A NEW CRITERION FOR INTEGRAL MODULAR CATEGORIFICATION

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Abstract. This paper establishes a highly efficient new necessary criterion for integral modular categorification, which allows us to complete the classification of categorifiable integral modular data up to rank 14, and up to rank 25 in the odd-dimensional case.

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1. Introduction

The classification of integral modular fusion categories is a central and profoundly challenging problem in the theory of tensor categories, with deep connections to representation theory, conformal field theory, and topological quantum computation. Although a complete classification remains out of reach, significant progress has been recently achieved in low ranks [1, 3, 5, 6, 15]. The main result of Czenky *et al.* [5, 6], completed in [1, 12], is the classification of all odd-dimensional modular fusion categories of rank less than 25. Furthermore, they proved in [6] that every non-pointed and non-perfect modular fusion category of rank 25 is equivalent to some $\mathcal{Z}(\mathrm{Vec}(C_7 \rtimes C_3, \omega))$. The principal result of Alekseyev *et al.* [1] is the classification of all categorifiable integral modular data up to rank 13. They provided exhaustive lists of candidate types for rank 14, and for rank 25 in the odd-dimensional case, though many of these remained unresolved.

This paper provides the decisive breakthrough needed to complete this classification. We introduce a powerful new necessary criterion that generalizes a key argument from the proof of [10, Lemma 9.3]. It proves remarkably effective, allowing us to eliminate the vast majority of the remaining candidate types.

The core of our result is Theorem 1.1, which bounds the global FPdim of the fusion category in terms of the FPdims of carefully chosen pairs of simple objects.

Theorem 1.1. *Let $\mathcal{C}$ be an integral modular fusion category. For any non-invertible simple object $X \in \mathcal{C}_{ad}$ and any prime p dividing $\mathrm{FPdim}(X)$ that is coprime to $\mathrm{FPdim}(\mathcal{C}_{pt})$, there exists a non-invertible simple object $Y \in \mathcal{C}$ with p coprime to $\mathrm{FPdim}(Y)$ such that*

$$\mathrm{lcm}(\mathrm{FPdim}(X), \mathrm{FPdim}(Y))^2 + \mathrm{FPdim}(X)^2 \mathrm{FPdim}(\mathcal{C}_{pt}) \leq \mathrm{FPdim}(\mathcal{C}).$$

In the perfect case—i.e., when $\mathcal{C}_{pt}$ is trivial and consequently $\mathcal{C}_{ad} = \mathcal{C}$—Theorem 1.1 specializes as follows:

Corollary 1.2. *Let $\mathcal{C}$ be a perfect integral modular fusion category. For any non-invertible simple object $X \in \mathcal{C}$ and any prime p dividing $\mathrm{FPdim}(X)$, there exists a non-invertible simple object $Y \in \mathcal{C}$ with p coprime to $\mathrm{FPdim}(Y)$ such that*

$$\mathrm{lcm}(\mathrm{FPdim}(X), \mathrm{FPdim}(Y))^2 + \mathrm{FPdim}(X)^2 \leq \mathrm{FPdim}(\mathcal{C}).$$

The inequality in Theorem 1.1 is a natural consequence of applying Galois automorphisms to the column orthogonality and norm identities of the S-matrix. In §2, we provide a brief overview of modular fusion categories and develop the Galois-theoretic preliminaries necessary for the proof of Theorem 1.1, which is presented in §3. Theorem 1.1 has been fully implemented in SageMath [17], see §5, directly automating the step-by-step procedure used in the hand proof of Lemma 4.2.

The strength of our criterion is immediately demonstrated by two major corollaries proved in §4:

Corollary 1.3. *Every non-pointed integral modular fusion category of rank 14 is equivalent to some $\mathcal{Z}(\mathrm{Vec}(A_4, \omega))$.*

In particular, in connection with [1, Question 1.8], [10, Question 2], and [13, Question 1.3], any non-pointed simple integral modular fusion category, if it exists, must have rank at least 15. Moreover, in the odd-dimensional case, it must have rank at least 27, since:

Corollary 1.4. *Every non-pointed odd-dimensional modular fusion category of rank 25 is equivalent to some $\mathcal{Z}(\mathrm{Vec}(C_7 \rtimes C_3, \omega))$.*

Moreover, we demonstrate the continuing effectiveness of our method by drastically reducing the number of unresolved cases at rank 15, mentioned in [1, §11], from 9027 to just 2481 types. This reduction offers a clear and manageable path forward for the ongoing classification program.

2. Preliminaries

2.1. **Modular fusion categories.** We briefly recall standard definitions and properties regarding modular fusion categories. We refer the reader to [9] for a comprehensive treatment of this subject.

A *fusion category* $\mathcal{C}$ over $\mathbb{C}$ is a $\mathbb{C}$-linear, semisimple, rigid monoidal category with finitely many isomorphism classes of simple objects, finite-dimensional Hom spaces, and a simple unit object $\mathbf{1}$. The set of isomorphism classes of simple objects is denoted by $\mathcal{O}(\mathcal{C})$.

The *Frobenius-Perron dimension* of a simple object $X \in \mathcal{O}(\mathcal{C})$, denoted $\mathrm{FPdim}(X)$, is defined as the Frobenius-Perron eigenvalue of the matrix representing left multiplication by the class of X in the Grothendieck ring of $\mathcal{C}$. The Frobenius-Perron dimension of the category itself is defined as $\mathrm{FPdim}(\mathcal{C}) = \sum_{X \in \mathcal{O}(\mathcal{C})} \mathrm{FPdim}(X)^2$. A fusion category $\mathcal{C}$ is called *integral* if $\mathrm{FPdim}(X) \in \mathbb{Z}$ for all $X \in \mathcal{O}(\mathcal{C})$.

A *braided fusion category* is a fusion category equipped with a natural isomorphism $c_{X,Y} : X \otimes Y \xrightarrow{\sim} Y \otimes X$ satisfying the hexagon axioms. A braided fusion category equipped with a *spherical* structure is called a *pre-modular* category. The S-matrix of a pre-modular category is the square matrix $S = (s_{X,Y})_{X,Y \in \mathcal{O}(\mathcal{C})}$ defined by the categorical trace of the double braiding $c_{Y,X} \circ c_{X,Y}$. A pre-modular category is *modular* if its S-matrix is non-degenerate.

2.2. **Galois theory in modular categories.** Galois theory provides a powerful toolkit for studying modular fusion categories. The entries $s_{X,Y}$ of the S-matrix are cyclotomic integers, inherently lying in a cyclotomic field $\mathbb{Q}(\zeta_n)$ for some integer n.

A crucial property, which forms the basis of the criterion we develop, is that the ratios $\frac{s_{X,Y}}{\mathrm{FPdim}(X)}$ are algebraic integers [9, Theorem 8.13.11]. Because the Galois group $\mathrm{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q})$ acts on the set of simple objects and commutes with complex conjugation, applying Galois automorphisms to structural identities—such as the column orthogonality of the S-matrix—allows one to extract strong arithmetic and numerical constraints on the dimensions of simple objects. This approach generalizes techniques successfully employed in [10, Lemma 9.3].

2.3. **Properties of algebraic integers.**

Definition 2.1. An algebraic integer x is called *totally positive* if all of its Galois conjugates are positive.

Definition 2.2. A *cyclotomic integer* is an algebraic integer that lies in a cyclotomic field $\mathbb{Q}(\zeta_n)$ for some n, where $\zeta_n = e^{2\pi i/n}$. Equivalently, a cyclotomic integer is an element of $\mathbb{Z}[\zeta_n]$ for some n.

Theorem 2.3. *Let x be a nonzero cyclotomic integer and let $m \in \mathbb{Z}_{>0}$. If $\frac{x}{m}$ is an algebraic integer, then there exists a Galois conjugate y of x such that $|y| \geq m$.*

Proof. We prove the theorem by reducing to some elementary lemmas:

Lemma 2.4. *Let K be the cyclotomic field $\mathbb{Q}(\zeta_n)$. For every $\sigma \in \mathrm{Gal}(K/\mathbb{Q})$ and every $x \in K$ we have*

$$\sigma(\overline{x}) = \overline{\sigma(x)}.$$

Equivalently, complex conjugation commutes with every element of $\mathrm{Gal}(K/\mathbb{Q})$.

Proof. By the Kronecker–Weber theorem, $\mathrm{Gal}(K/\mathbb{Q})$ is Abelian. Hence, the complex conjugation $\zeta_n \mapsto \zeta_n^{-1}$, viewed as an element of this group, is necessarily central. The result follows. ☐

Lemma 2.5. *Let y be a nonzero cyclotomic integer. Then $z = y\overline{y} = |y|^2$ is a totally positive algebraic integer.*

Proof. Since y is an algebraic integer, so is its complex conjugate $\overline{y}$; hence $z = y\overline{y}$ is an algebraic integer.

By Lemma 2.4, for each Galois automorphism σ we have

$$\sigma(z) = \sigma(y)\,\overline{\sigma(y)} = |\sigma(y)|^2.$$

Each value $|\sigma(y)|^2$ is positive (since $y \neq 0$). Thus all conjugates of z are positive, i.e. z is totally positive. ☐

Lemma 2.6. *Let x be a totally positive algebraic integer and let $m \in \mathbb{Z}_{>0}$. If $\frac{x}{m}$ is an algebraic integer, then there exists a conjugate y of x such that $y \geq m$.*

Proof. Let $x_1, \ldots, x_d$ be the (positive) conjugates of x. Since $\frac{x}{m}$ is an algebraic integer, its norm

$$N\Big(\frac{x}{m}\Big) = \prod_{i=1}^{d} \frac{x_i}{m} = \frac{N(x)}{m^d}$$

is a nonzero integer. If all conjugates satisfied $x_i < m$, then each factor x_i/m lies in $(0,1)$, giving

$$0 < \prod_{i=1}^{d} \frac{x_i}{m} < 1,$$

a contradiction. Therefore some conjugate x_i satisfies $x_i \geq m$. ☐

We now complete the proof of Theorem 2.3. By Lemma 2.5, $z := |x|^2$ is totally positive, while $z/m^2 = |x/m|^2$ is an algebraic integer. By Lemma 2.6, z has a conjugate $z_i \geq m^2$. Since the conjugates of z are exactly $|\sigma(x)|^2$, some conjugate $y = \sigma(x)$ satisfies $|y|^2 \geq m^2$, i.e. $|y| \geq m$. ☐

Observe that Theorem 2.3 extends to all algebraic integers x such that $\sigma(\overline{x}) = \overline{\sigma(x)}$ for all σ.

Lemma 2.7. *Let x be an algebraic integer and let $m, n \in \mathbb{Z} \setminus \{0\}$. If both x/m and x/n are algebraic integers, then so is $x/\operatorname{lcm}(m,n)$.*

Proof. Let $g = \gcd(m,n)$. By Bézout's identity, there exist integers u, v such that $um + vn = g$. Since the set of algebraic integers is a $\mathbb{Z}$-module, any integral linear combination of algebraic integers is again an algebraic integer. In particular,

$$u\frac{x}{m} + v\frac{x}{n} = x\Big(\frac{u}{m} + \frac{v}{n}\Big) = x \cdot \frac{un + vm}{mn} = x \cdot \frac{g}{mn}$$

is an algebraic integer. Noting that

$$\frac{g}{mn} = \pm\frac{1}{\operatorname{lcm}(m,n)},$$

we deduce that $x/\operatorname{lcm}(m,n)$ is also an algebraic integer. ☐

Remark 2.8. As noted by a referee, the bound in Lemma 2.6 can be strengthened using Siegel's result [18, Theorem III], which states that the trace of a totally positive algebraic integer $\alpha \neq 1$ of degree n is at least $\frac{3}{2}n$, so that at least one conjugate β of α satisfies $\beta \geq \frac{3}{2}$. Applied to x/m, this shows that at least one conjugate y of x satisfies $y \geq \frac{3}{2}m$ whenever $x \neq m$. Siegel's proof requires several pages of intricate analysis, whereas our elementary, few-line argument yielding $y \geq m$ suffices for our criterion. Nevertheless, this stronger estimate suggests a potential avenue for refining Theorem 1.1 in future classification work, though this would necessitate handling the case $|s_{X,Y}| = \operatorname{lcm}(\mathrm{FPdim}(X), \mathrm{FPdim}(Y))$.

3. Proof of Theorem 1.1

Proof. Consider the orthogonality relation between the columns $(s_{X,Y})_Y$ and $(s_{1,Y})_Y = (\mathrm{FPdim}(Y))_Y$ of the S-matrix ([9, Proposition 8.14.2]):

$$\sum_{Y\in\mathcal{O}(\mathcal{C})} \frac{s_{X,Y}}{\mathrm{FPdim}(X)}\mathrm{FPdim}(Y) = 0.$$

There exists a non-invertible simple object Y_1 such that p is coprime to $\mathrm{FPdim}(Y_1)$ and $s_{X,Y_1} \neq 0$. Indeed, otherwise the only possible nonzero summands would come from the invertible objects Y, for which $s_{X,Y} = \mathrm{FPdim}(X)$ since $X \in \mathcal{C}_{ad} = \mathcal{C}'_{pt}$ by [9, Corollary 8.22.8], and from simple objects Y with $p \mid \mathrm{FPdim}(Y)$. As $\frac{s_{X,Y}}{\mathrm{FPdim}(X)}$ is an algebraic integer by [9, Theorem 8.13.11(ii)], this implies that

$$\sum_{Y\in\mathcal{O}(\mathcal{C})} \frac{s_{X,Y}}{\mathrm{FPdim}(X)}\mathrm{FPdim}(Y) \;\equiv\; \mathrm{FPdim}(\mathcal{C}_{pt}) \pmod p,$$

a contradiction, since p is coprime to $\mathrm{FPdim}(\mathcal{C}_{pt})$. Since S is symmetric, both ratios

$$\frac{s_{X,Y_1}}{\mathrm{FPdim}(X)} \quad \text{and} \quad \frac{s_{X,Y_1}}{\mathrm{FPdim}(Y_1)}$$

are algebraic integers. Hence, by Lemma 2.7

$$\frac{s_{X,Y_1}}{\mathrm{lcm}(\mathrm{FPdim}(X),\mathrm{FPdim}(Y_1))}$$

is also an algebraic integer. In particular, $\frac{s_{X,Y_1}}{\mathrm{FPdim}(X)}$ is divisible by

$$M := \frac{\mathrm{lcm}(\mathrm{FPdim}(X),\mathrm{FPdim}(Y_1))}{\mathrm{FPdim}(X)}.$$

Next, consider the column norm identity ([9, Proposition 8.14.2]):

$$\sum_{Y\in\mathcal{O}(\mathcal{C})} \left|\frac{s_{X,Y}}{\mathrm{FPdim}(X)}\right|^2 = \frac{\mathrm{FPdim}(\mathcal{C})}{\mathrm{FPdim}(X)^2}. \tag{3.1}$$

Since $\mathcal{C}$ is integral, the right-hand side of (3.1) is rational and hence invariant under any Galois automorphism. By [9, Theorem 8.13.11(ii) and Theorem 8.14.7] and Lemma 2.5, each summand on the left-hand side is a totally positive cyclotomic integer. Each term corresponding to an invertible Y equals 1, while

$$x := \left|\frac{s_{X,Y_1}}{\mathrm{FPdim}(X)}\right|^2$$

is a totally positive cyclotomic integer divisible by M^2. By Lemma 2.6, there exists a Galois automorphism σ such that $\sigma(x) \geq M^2$. Applying σ to (3.1) yields

$$\frac{\mathrm{FPdim}(\mathcal{C})}{\mathrm{FPdim}(X)^2} \geq \mathrm{FPdim}(\mathcal{C}_{pt}) + M^2.$$

Thus

$$\frac{\mathrm{FPdim}(\mathcal{C})}{\mathrm{FPdim}(X)^2} \geq \mathrm{FPdim}(\mathcal{C}_{pt}) + \left(\frac{\mathrm{lcm}(\mathrm{FPdim}(X),\mathrm{FPdim}(Y_1))}{\mathrm{FPdim}(X)}\right)^2,$$

The result follows by multiplying both sides by $\mathrm{FPdim}(X)^2$. □

4. Applications

4.1. **Proof of Corollary 1.3.** Let us divide the proof into the non-perfect and the perfect cases:

Proposition 4.1. *A non-pointed non-perfect integral modular fusion category of rank* 14 *is equivalent to* $\mathcal{Z}(\mathrm{Vec}(A_4,\omega))$.

Proof. By [8, Theorem 4.64] and [1, Proposition 11.1], we are reduced to exclude the following seven types:

```
L=[[1,1,2,3,3,24,24,42,42,56,56,56,84,84],
[1,1,2,3,3,24,120,150,150,200,200,200,300,300],
[1,1,24,24,36,40,45,45,90,90,90,180,180,180],
[1,1,40,84,90,126,315,315,504,630,840,1260,1260,1260],
[1,1,45,45,90,140,168,168,630,630,840,1260,1260,1260],
[1,1,60,60,84,140,189,189,540,1260,1260,1890,1890,1890],
[1,1,90,90,90,108,140,378,945,945,1260,1890,1890,1890]]
```

All but the first two types are excluded by Theorem 1.1. The third type is ruled out as follows by hand.

Lemma 4.2. *There is no modular fusion category of type* $t = [1, 1, 24, 24, 36, 40, 45, 45, 90, 90, 90, 180, 180, 180]$.

Proof. Let $\mathcal{C}$ be a modular fusion category. By [8, Proposition 2.3 and Lemma 3.31], the adjoint subcategory $\mathcal{C}_{ad}$ is the neutral component of the universal grading, which is a faithful grading by the group G of invertible objects. Hence, by [9, Theorem 3.5.2], $\mathrm{FPdim}(\mathcal{C}_{ad}) = \mathrm{FPdim}(\mathcal{C})/|G|$.

Assume that $\mathcal{C}$ has type t. Then $|G| = 2$ and $\mathcal{C}_{ad}$ must have type $[1, 1, 24, 24, 36, 40, 45, 45, 90, 90, 90, 180]$, as this is the only possible type for a fusion subcategory of dimension $\mathrm{FPdim}(\mathcal{C})/2$. Indeed, the sum of the squares of all entries of t smaller than 180 equals exactly 180^2, so any sublist containing 1 whose squares sum to half the total $4 \cdot 180^2$ must consist of all these smaller entries together with exactly one 180.

Let $X \in \mathcal{O}(\mathcal{C})$ be a non-invertible simple object with $\mathrm{FPdim}(X) = 36$. Take $p = 3$, a prime divisor of $\mathrm{FPdim}(X)$ that is coprime to $\mathrm{FPdim}(\mathcal{C}_{pt}) = 2$. Any non-invertible object $Y \in \mathcal{O}(\mathcal{C})$ whose Frobenius–Perron dimension is coprime to $p = 3$ must satisfy $\mathrm{FPdim}(Y) = 40$. However,

$$\begin{aligned}&\mathrm{lcm}(\mathrm{FPdim}(X), \mathrm{FPdim}(Y))^2 + \mathrm{FPdim}(X)^2\mathrm{FPdim}(\mathcal{C}_{pt})\\ =\,&\mathrm{lcm}(36, 40)^2 + 36^2 \times 2 = 132192 > 129600 = \mathrm{FPdim}(\mathcal{C}),\end{aligned}$$

which contradicts Theorem 1.1. □

The `NewCrit` function from §5 automates the exclusions by Theorem 1.1, directly following the procedure in the proof of Lemma 4.2. The computation below confirms that all remaining types are excluded, except for the first two.

```
sage: for l in L:
....:     if NewCrit(l):
....:         print(l)
[1,1,2,3,3,24,24,42,42,56,56,56,84,84]
[1,1,2,3,3,24,120,150,150,200,200,200,300,300]
```

Assume $\mathcal{C}$ is a modular fusion category of either type. The simple objects with FPdims in the common sublist $[1, 1, 2, 3, 3]$ are closed under tensor products—and thus form a fusion subcategory $\mathcal{D}$ of type $[1, 1, 2, 3, 3]$—because their maximum squared FPdim ($3^2 = 9$) is strictly less than the smallest remaining FPdim (24).

Lemma 4.3. *The fusion subcategory* $\mathcal{D}$ *is maximal Tannakian, and isomorphic to* $\mathrm{Rep}(S_4)$.

Proof. By [14, Theorem 3.2] or [8, Theorem 3.14],

$$\mathrm{FPdim}(\mathcal{D})\,\mathrm{FPdim}(\mathcal{D}') = \mathrm{FPdim}(\mathcal{C}). \tag{4.1}$$

In each case, there exists a unique possible type for a fusion subcategory of dimension $\mathrm{FPdim}(\mathcal{C})/\mathrm{FPdim}(\mathcal{D})$: it is $[1, 1, 2, 3, 3, 24, 24]$ in the first case and $[1, 1, 2, 3, 3, 24, 120]$ in the second. Consequently, $\mathcal{D}'$ must be of that type, and hence has rank 7. In particular, $\mathcal{D} \cap \mathcal{D}' = \mathcal{D}$, hence $\mathcal{D}$ is symmetric. According to the classification of premodular categories of rank 5 in [2, Theorem I.1], a symmetric fusion category of type $[1, 1, 2, 3, 3]$ is Tannakian and isomorphic to $\mathrm{Rep}(S_4)$.

Regarding maximality, assume that $\mathcal{E}$ is a Tannakian subcategory properly containing $\mathcal{D}$. Then $\mathcal{E}'$ is properly contained in $\mathcal{D}'$ (since, by [14, Theorem 3.2], we have $\mathcal{E}'' = \mathcal{E}$ and $\mathcal{D}'' = \mathcal{D}$ since $\mathcal{C}$ is modular). If $\mathcal{E}' = \mathcal{E}$, then $\mathcal{E}$ is Lagrangian, and hence $\mathrm{FPdim}(\mathcal{C}) = \mathrm{FPdim}(\mathcal{E})^2$; see [8, Definition 4.57 and Theorem 4.64]. The only possible case would then be that $\mathcal{E}$ has type $[1, 1, 2, 3, 3, 24]$, but a direct computation in GAP [11] shows that there is no finite group with these character degrees:

```
gap> Filtered(AllSmallGroups(600),
     g -> SortedList(CharacterDegrees(g)) = [[1,2],[2,1],[3,2],[24,1]]);
[  ]
```

Therefore, we obtain the chain of proper inclusions

$$\mathcal{D} \subsetneq \mathcal{E} \subsetneq \mathcal{E}' \subsetneq \mathcal{D}',$$

a contradiction, since $\mathcal{D}'$ consists of $\mathcal{D}$ together with only two additional simple objects. □

By [8, Corollary 5.15], the core of $\mathcal{C}$ (that is, the de-equivariantization of $\mathcal{D}'$) is an integral modular fusion category, and by (4.1) and [8, Proposition 4.26], its global dimension is $\mathrm{FPdim}(\mathcal{C})/\mathrm{FPdim}(\mathcal{D})^2 = 7^2$ or 5^4, respectively. However, an integral modular fusion category of global dimension p^2, for a prime p, is necessarily pointed, as it is half-Frobenius [9, Proposition 8.14.6]. It is also pointed when the global dimension equals p^4, by [7, Lemma 4.11]. Hence the rank-7 fusion category $\mathcal{D}'$ must be an S_4-equivariantization of a pointed fusion category of rank 7^2 or 5^4, respectively. We now show that this is impossible.

Lemma 4.4. *Let G and H be finite groups. Then the rank of the G-equivariantization* $\mathrm{Vec}(H,\omega)^G$ *is at least the number of orbits of the corresponding group automorphism action of G on H.*

Proof. Immediate from [4, Corollary 2.13]. □

A group automorphism action of G on H is a group homomorphism from G to $\mathrm{Aut}(H)$, hence an isomorphism between a quotient of G and a subgroup of $\mathrm{Aut}(H)$. We verified with GAP [11] that the quotients of S_4 are isomorphic to C_1, C_2, S_3, and S_4:

```
gap> List(NormalSubgroups(SymmetricGroup(4)),
     H -> StructureDescription(FactorGroup(SymmetricGroup(4), H)));
[ "1", "C2", "S3", "S4" ]
```

and that, if $|H| = 7^2$ (so H Abelian), then $\mathrm{Aut}(H)$ has no subgroup isomorphic to S_4:

```
gap> List(AllSmallGroups(49), g -> Filtered(
     ConjugacyClassesSubgroups(AutomorphismGroup(g)),
     c -> StructureDescription(Representative(c)) = "S4"));
[ [  ], [  ] ]
```

Since the order of a proper quotient of S_4 is at most $|S_3| = 6$, the number of orbits of such a group acting on a group of order 49 is at least $49/6 > 7$. The case $|H| = 5^4$ is even simpler: the number of orbits of a group of order at most 24 acting on a group of order 625 is at least $625/24 > 7$. Both $49/6$ and $625/24$ are strictly greater than 7, the rank of $\mathcal{D}'$. Therefore, by Lemma 4.4, $\mathcal{D}'$ cannot be an S_4-equivariantization of a pointed fusion category of rank 7^2 or 5^4. This contradiction excludes the two remaining cases. □

Proposition 4.5. *There is no non-trivial perfect integral modular fusion categories up to rank* 14.

Proof. As mentioned in [1, §11], there remain 27 types to consider, all of which are excluded by Theorem 1.1, as confirmed by the following computation:

```
LL=[[1,30,35,63,90,90,126,140,252,315,420,630,630,630],
[1,30,30,30,105,105,105,120,140,168,280,420,420,420],
[1,35,60,105,105,168,210,240,560,560,560,560,840,840],
[1,35,84,108,135,140,252,315,420,1260,1260,1890,1890,1890],
[1,35,105,108,126,135,140,378,420,1260,1260,1890,1890,1890],
[1,40,105,105,168,175,200,350,1050,1050,1400,2100,2100,2100],
[1,45,56,63,63,70,120,120,360,840,840,1260,1260,1260],
[1,50,50,105,105,168,175,210,600,1400,1400,2100,2100,2100],
[1,60,105,150,168,175,200,280,525,1400,1400,2100,2100,2100],
[1,63,105,140,189,280,360,378,1080,2520,2520,3780,3780,3780],
[1,63,63,70,70,180,270,630,756,945,1260,1890,1890,1890],
[1,70,75,84,84,84,84,150,525,525,700,1050,1050,1050],
[1,70,75,105,140,150,168,350,525,1400,1400,2100,2100,2100],
```

```
[1,70,75,150,168,175,210,280,525,1400,1400,2100,2100,2100],
[1,70,75,75,168,200,200,300,525,1400,1400,2100,2100,2100],
[1,75,105,105,150,168,210,350,1050,1050,1400,2100,2100,2100],
[1,75,140,150,168,175,175,420,420,1400,1400,2100,2100,2100],
[1,245,270,270,675,882,2450,11025,18900,44100,44100,66150,66150,66150],
[1,245,675,882,2450,2700,11025,13230,13230,44100,44100,66150,66150,66150],
[1,270,1225,2025,2268,4900,5670,33075,56700,132300,132300,198450,198450,198450],
[1,315,490,810,980,1620,2268,8820,19845,19845,26460,39690,39690,39690],
[1,315,875,882,1125,1960,18000,73500,126000,294000,294000,441000,441000,441000],
[1,350,405,1750,2268,3375,3780,23625,40500,94500,94500,141750,141750,141750],
[1,350,945,1620,1750,2268,4725,23625,40500,94500,94500,141750,141750,141750],
[1,441,675,1372,2700,3430,18522,77175,132300,308700,308700,463050,463050,463050],
[1,490,675,882,1225,2700,6615,14700,14700,44100,44100,66150,66150,66150],
[1,945,1225,1620,2268,4900,40500,165375,283500,661500,661500,992250,992250,992250]]
sage: for t in LL:
....:     if NewCrit(t):
....:         print(t)
sage:       # all excluded!
```

□

4.2. **Proof of Corollary 1.4.** According to [1, §12], completing the classification of the modular data of odd-dimensional modular fusion categories of rank 25 requires addressing the perfect case. This is precisely what the following proposition achieves.

Proposition 4.6. *There is no perfect odd-dimensional modular fusion category of rank* 25*.*

Proof. By [1, Theorem 12.8], there remains to consider three possible types, all excluded by Theorem 1.1:

```
sage: L=[[[1,1],[75,2],[91,4],[175,2],[585,2],[975,2],[2275,2],[4095,2],[6825,8]],
....: [[1,1],[75,2],[91,4],[175,2],[975,2],[2275,2],[2925,4],[6825,8]],
....: [[1,1],[135,4],[165,2],[189,2],[315,2],[385,2],[1155,2],[2079,2],[3465,8]]]
sage: format = lambda T: [t[0] for t in T for i in range(t[1])]
sage: for T in L:
....:     l=format(T)
....:     if NewCrit(l):
....:         print(l)
sage:       # all excluded!
```

□

Remark 4.7. Among the 91 possible types mentioned in the proof of [1, Theorem 12.8], all but three can be directly ruled out by Theorem 1.1, while all but fifteen are excluded by [1, Theorem 8.7]. Taken together, these two theorems rule out all 91 possible types.

4.3. **Discussion about rank 15.** In the rank-15 case considered in [1, §11], 9027 types remained, comprising 399 non-perfect and 8628 perfect types. Applying Theorem 1.1 reduced this set to 2481 types, including 140 non-perfect and 2341 perfect. They are available at `Rank15/RemainingTypes.txt` in the repository [16]. A few examples are:

- Few non-perfect ones:

  ```
  [1,1,1,1,4,4,12,12,36,36,108,108,162,162,162]
  [1,1,1,3,12,12,60,60,100,100,100,100,100,100,150]
  [1,1,135,135,140,252,1080,1512,1890,5670,5670,7560,11340,11340,11340]
  ```

- Few perfect ones:

  ```
  [1,20,20,21,21,35,40,56,70,84,280,280,420,420,420]
  [1,21,21,21,24,24,60,140,168,210,280,420,420,420,840]
  [1,24,24,56,60,70,105,315,756,945,2520,2520,3780,3780,3780]
  [1,25,25,28,28,60,84,280,350,525,1400,1400,2100,2100,2100]
  [1,27,27,30,35,40,180,360,504,1080,2520,2520,3780,3780,3780]
  [1,28,28,30,35,80,84,105,105,420,420,560,840,840,840]
  ```

```
[1,30,30,30,105,105,105,120,140,168,280,420,420,420,840]
[1,32,35,42,45,45,120,144,840,1440,3360,3360,5040,5040,5040]
[1,35,35,35,42,96,112,360,1120,2520,2520,3360,5040,5040,5040]
[1,36,36,45,70,70,189,270,756,945,2520,2520,3780,3780,3780]
[1,40,42,42,45,105,280,315,336,630,1680,1680,2520,2520,2520]
[1,42,42,60,175,189,280,2700,3150,4725,12600,12600,18900,18900,18900]
[1,45,45,45,56,70,420,504,756,756,2520,2520,3780,3780,3780]
[1,48,63,126,175,240,315,1800,2520,2800,8400,8400,12600,12600,12600]
[1,49,49,56,175,250,294,3000,12250,21000,49000,49000,73500,73500,73500]
```

5. SageMath codes

To ensure complete transparency and independent verifiability, we provide below a fully self-contained SageMath code. The proof of Lemma 4.2 shows in detail, by hand, how to exclude a type using Theorem 1.1, and this code directly automates that procedure. By running it on the provided lists of candidate types, readers can readily reproduce the exclusions discussed in §4. The auxiliary function `ModularPartitions` is described in detail in [1, §8.1].

```
def NewCrit(l):
 pt = l.count(1)
 spt=set(Integer(pt).prime_factors())
 d=sum(i**2 for i in l)
 S=list(set(l))
 S.sort()
 M=[]
 MP=ModularPartitions(l)
 for P in MP:
  S0=list(set(P[0]))
  S0.sort()
  if NewCritInter(S0,S,spt,pt,d):
   M.append(P)
 if len(M)>0 and len(M)<len(MP) and pt>1:
  print('only need to consider the partitions in ', M)
 return len(M)>0

def NewCritInter(S0,S,spt,pt,d):
 for i in S0[1:]:
  for p in set(Integer(i).prime_factors())-spt:
   c=0
   for j in S[1:]:
    if j%p!=0 and lcm(i,j)**2 + pt*(i**2) <= d:
     c=1
     break
   if c==0:
    return False
 return True

# generate modular partitions of type l
def ModularPartitions(l):
    d = sum(i^2 for i in l)
    p = l.count(1)
    # assert d % p == 0
    if d % p != 0:
        return []
```

```
    U = sorted(set(l))  # unique elements in l
    S = [[p.count(u^2) for u in U] for p in gen_mparts([i^2 for i in reversed(l)], d//p)]
    return sorted(sorted(sum(([u] * q for u, q in zip(U, Qi)), [])
            for Qi in Q) for Q in VectorPartitions([l.count(u) for u in U], parts=S))

# generate submultisets of list M (sorted in nonincreasing order) with a given sum s
def gen_mparts(M,s,i=0):
    if s==0:
        yield tuple()
        return
    while i<len(M) and M[i]>s:
        i += 1
    prev = 0
    while i<len(M):
        if M[i]!=prev:
            for p in gen_mparts(M,s-M[i],i+1):
                yield p+(M[i],)
        prev = M[i]
        i += 1
```

Let us apply `NewCrit` to the rank-22 type of the Drinfeld center $\mathcal{Z}(\mathrm{Rep}(A_5))$ and a rank-14 type:

```
sage: l1=[1,3,3,4,5,12,12,12,12,12,12,12,12,12,12,15,15,15,15,20,20,20]
sage: NewCrit(l1)
True
sage: l2=[1,30,35,63,90,90,126,140,252,315,420,630,630,630]
sage: NewCrit(l2)
False
```

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